

# The Operator's Manual

*How to run, configure and operate the BSS — by role and by task. The companion volume, *The Honest Machine*, tells how it was built and why; this book tells you which button, which endpoint, which environment variable. Everything here is proven by the suite named beside it.*

## Contents

- |                               |                             |
|-------------------------------|-----------------------------|
| 1. Getting started            | 7. The host administrator   |
| 2. Surfaces & sign-ins        | 8. AI configuration         |
| 3. The product owner          | 9. The seam catalog         |
| 4. The CSR                    | 10. Environment reference   |
| 5. Billing & money operations | 11. Extension API reference |
| 6. Partner channels           | 12. The mock integrations   |

## 1 • Getting started

```
docker compose up -d          # ~90 containers; first build takes a while
open http://localhost:8080/shop/ # the storefront (guest browsing works)
```

The gateway is :8080 ; Keycloak is :8085 (admin/admin). Two seed operators run side by side: **genalpha** (EN/EUR, realm `bss` ) and **nova** (NO/NOK, realm `nova` ) — plus any operator you mint (§7). Demo card `4242 4242 4242 4242` pays; anything ending `0002` declines. Promo code `WELCOME10` . Serviceable fiber postcodes start `111 / 222 / 333` .

**The one discipline:** every feature in this manual has an end-to-end Playwright suite in `ops/e2e/` . When in doubt about how something behaves, the suite is the specification: `node ops/e2e/<name>_test.js` runs it against the live stack.

## 2 • Surfaces & sign-ins

| URL                                         | WHO                                      | DEMO SIGN-IN                                                                           |
|---------------------------------------------|------------------------------------------|----------------------------------------------------------------------------------------|
| <code>/shop/</code>                         | Customers (self-register or guest)       | <code>kai@bss.local</code> / <code>kai</code>                                          |
| <code>/app/</code>                          | Customers on mobile (Expo web)           | <code>emil@acme.example</code> / <code>emil</code>                                     |
| <code>/biz/</code>                          | B2B org admins + invited members         | <code>bianca@acme.example</code> / <code>bianca</code>                                 |
| <code>/csr/</code>                          | Agents (role-scoped powers)              | <code>agent-anna</code> / <code>agent ; junior: jo@bss.local</code> / <code>jo</code>  |
| <code>/console/</code>                      | Back office (role-scoped tabs)           | <code>demo</code> / <code>demo ; product-only: pat@bss.local</code> / <code>pat</code> |
| <code>/dealer-app/</code>                   | Retail-partner clerks & telesales agents | any account whose org holds a dealer agreement                                         |
| <code>shop.nova.localhost:8080/shop/</code> | Nova's white-label storefront            | <code>nils@nova.local</code> / <code>nils</code>                                       |
| <code>biz.nova.localhost:8080/biz/</code>   | Nova's B2B console (Norwegian)           | <code>birgit@fjellheim.no</code> / <code>birgit</code>                                 |

White-labeling is hostname-driven: `shop.<tenant>.localhost` serves that operator's brand, locale and currency from the live registry (§7).

## 3 • The product owner

### Catalog (console → Product Offerings / Specifications / Prices)

Offerings are TMF620 data: hard and soft bundles ( `bundledProductOfferingOption` cardinality, pick-N-of-M choice groups), characteristic-conditioned prices (" +2.00/mo when colour = Titanium"), commitment terms, categories that drive placement *and* fulfilment (Top-ups and Insurance are billing-only; Partner services mint entitlement codes; everything else draws an MSISDN + SIM). Artwork rides the offering's attachments — internal store or the tenant's own PIM.

### Rules & pricing as data (console → Rules)

Order rules and dynamic pricing are JSON-logic rows with a dry-run panel. A price change is a rule, not a release. See `docs/product-rules.md` for the authoring guide.

## Product Copilot (console → Copilot)

Chat a product into the catalog: the model proposes specs/prices/offerings as TMF620 payloads, the console shows a human card, and a deterministic executor applies it with *your* token on confirmation. The model never writes.

## Product Advisor (console → Product advisor) — suite #52

Findings with receipts: top-up attach counted from inventory ("2 of 2 customers on this plan also bought top-ups"), market price gaps from the tenant's market feed (10% materiality threshold — small gaps are deliberately silent), an LLM narrative that never invents a number. **Adopt as draft...** births an `In study` offering; nothing reaches the shop until you promote it.

## Campaigns & growth (console → Campaigns / Journeys / Audiences)

Segments form from consented shop behavior; campaigns prove lift against real holdouts; journeys read the customer before speaking; A/B arms split deterministically; guardrails (frequency caps, quiet hours) are settings. Every message lands on the TMF683 timeline.

## 4 • The CSR

---

Search the 360 (multi-word, name-ranked), then act — every action logs itself on the customer's timeline:

| THE CUSTOMER ASKS                 | THE BUTTON / ACT                                                                                |
|-----------------------------------|-------------------------------------------------------------------------------------------------|
| "What's my PUK?" / "reset my PIN" | Reveal PUK · Reset PIN (owner notified — a silent credential change reads as fraud)             |
| "I lost my SIM"                   | Replace SIM (old card quarantined first)                                                        |
| "new number, please"              | Change number (old number quarantined with its story)                                           |
| "pause my subscription"           | Pause / Resume (charging pauses at the OCS; auto-resumes on the promised date)                  |
| "I can't pay all of it"           | Installments (parts sum exactly; miss one → reminder → grace → dunning)                         |
| "this charge is wrong"            | Dispute (collection pauses; resolution credits or upholds, in writing)                          |
| "send me my invoice"              | PDF (the exact document) · Email bill (to the address on file, never one dictated on the phone) |
| "my internet is slow"             | Diagnose (triage across usage/assurance/network facts)                                          |
| "give my number to my colleague"  | Transfer (three-way authorization; B2B handover keeps the company paying)                       |

Self-care mirrors most of these in the shop and app — customers act on their OWN line (ownership-checked; a foreign line answers 404, never 403).

## 5 • Billing & money operations

### The billing run

`POST /tmf-api/customerBillManagement/v4/billingRun` (staff). Segment-based and per-account: mid-month starts, mid-cycle plan changes and ceases each bill exactly their own days, labelled ("from the 16th, 15 days"). Payday alignment per customer ( `POST /individual/{id}/billingCycle` , day 1–28). The invariant the suites enforce from every direction: **no day is ever billed twice**.

### The bill as a document — suite #46

`GET /customerBill/{id}/document.pdf` (owner-scoped); `POST /customerBill/{id}/resend` emails it to the address on file. Delivery preference per customer: `paper | einvoice | digital` ( `POST /individual/{id}/billDelivery` ) — the customer picks the channel; the tenant keeps the plumbing.

## E-invoice formats are rows (console → Bill formats)

| CODE    | STANDARD                                  | SYNTAX             |
|---------|-------------------------------------------|--------------------|
| ehf     | EHF Billing 3.0 (Norway; carries the KID) | UBL 2.1            |
| peppol  | Peppol BIS Billing 3.0                    | UBL 2.1            |
| cii     | EN 16931                                  | UN/CEFACT CII      |
| aunz    | A-NZ Peppol BIS 3.0                       | UBL 2.1            |
| edifact | UN/EDIFACT INVOIC D96A                    | segments           |
| facturx | Factur-X / ZUGFeRD                        | PDF + embedded CII |

Adding a country is an INSERT (the suite added XRechnung through the form). The tenant's `BILL_DISTRIBUTION_FORMAT` picks a row by code; the renderer follows the row.

## Delivery, status, and the money coming home — suites #46/#47

**Deliveries** (console tab): the outbox-backed ledger — every bill's trip, attempts, the buyer's Peppol Invoice Response (accepted / rejected with their words / paid), one-click Retry on failures. **Remittance**: the bank posts camt.054, Nets OCR or BAI2 to `POST /bank/v1/remittance` (per-tenant `BANK_TOKEN`); a matched KID settles the bill through the card path's own guarantee; everything unclear parks on **Unapplied cash** (console tab) with its reason. Money is fail-closed: never guessed, never dropped.

## Dunning (console → Dunning)

The collections worklist: overdue installments, broken plans, what is still owed.

## 6 • Partner channels — suites #48/#51

**Being a dealer is a row.** Sign a chain: `POST /dealer/v1/agreement` (staff) with the org, commission per activation, and — for a chain with its own POS — the `OAuth2 clientId` that speaks for them. Clerks are org membership; foreign dealers see 404-shaped nothing.

## Retail (the CSP + external-retail model — think Elkjøp/Power)

**Counter sale**: sell against the customer's email; the order carries the dealer stamp. **Starter kits**: mint batches (code + SIM + attribution in the box); the customer self-activates ( `POST`

/dealer/v1/starterKit/activate ) and the line rides the SIM from the box. **POS API:** same endpoints, machine credential, per-partner rate limits at the edge (429 + Retry-After); the chain's own phone rides the sale as `device` context — never a billable item. **Commission:** pending → earned after the angreter window → clawed back with the reason if the customer leaves inside it.

## Telesales (outbound, shaped by the law)

**Dial list:** GET /dealer/v1/telesales/dialList?segment= — consented at the source (insight segments), every number DNC-washed, reserved citizens excluded *and counted*. **The call makes an OFFER** ( POST /dealer/v1/telesales/offer ): washed fail-closed (reserved refuses, unreachable register refuses, unconfigured tenant refuses); no order exists yet. **The customer confirms in writing** ( POST /telesales/v1/confirm , signed in, code from their inbox — or, for a cold prospect, after registering with the offered email); only then the order, the activation and the commission are born. Silence expires on a clock. Every dialer call logs to TMF683.

## 7 • The host administrator

---

### The tenant fleet is a file

infra/tenants/tenants.yml is the shared registry (mounted into all 31 services); every service live-refreshes it. NEW operators join the running fleet within one refresh interval; changes to a serving operator's seams/brand/hosts apply live; identity fields ( `id` , `issuer` , key endpoints) require a restart — by design.

### Minting an operator — suites #49/#50

```
# the script (~2 minutes to a billed customer):
ops/onboard-tenant.sh fjord "Fjord Mobil" da DKK "#1B6CA8"

# or the form: console → Operators → five fields → Save
# (host-tenant admins only; realm clone + registry block + seeded catalog;
# shop.<id>.localhost wears the brand the moment the gateway refreshes)
# live rebrand of a form-born operator:
PATCH /onboarding/v1/operator/{id} {"name","color","locale","currency"}
```

Seed operators (genalpha, nova) are protected from the form — their config is env.

### Staff & roles (console → Staff)

Grant/revoke whole areas per operator over the tenant's own IdP (TMF672) — no Keycloak console needed for daily administration.

## 8 • AI configuration — suite #53

---

Per tenant, all data: `ai-provider` ( `stub` — keyless CI default — `openai-compatible` , `anthropic` ), `ai-base-url` , `ai-api-key` , `ai-model` . **Tiered routing**: the task class lives at the call site — FAST (copywriting, summaries, narratives) vs SMART (product proposals, next-best-offer; the default when unclassified). Every value resolves tier-first:

```
AI_MODEL_FAST=swift-mini           # cheap model for volume work
AI_MODEL_SMART=frontier-x          # careful model for judgment
AI_PROVIDER_SMART=anthropic        # tiers go down to the PROVIDER:
AI_BASE_URL_SMART=https://api...   # a local endpoint for volume,
AI_API_KEY_SMART=...               # a frontier API for judgment — at once
```

## 9 • The seam catalog

---

Every external system is a per-tenant seam: configure it and the feature lights up; leave it blank and the feature degrades honestly (fail-open for enrichment, fail-closed for money, identity and consent).

| SEAM                      | TENANT FIELDS (ENV)                                   | BEHAVIOR WHEN BLANK                                       |
|---------------------------|-------------------------------------------------------|-----------------------------------------------------------|
| Email provider (ESP)      | DELIVERY_PROVIDER, ESP_URL, ESP_API_KEY, ESP_FROM     | in-app inbox only                                         |
| Bill distribution partner | BILL_DISTRIBUTION_{PROVIDER,URL,TOKEN,FORMAT,CHANNEL} | bills stay in-app/PDF                                     |
| Bank (remittance)         | BANK_TOKEN                                            | webhook refuses (money: fail-closed)                      |
| Do-not-call register      | DNC_URL, DNC_TOKEN                                    | outbound telesales refuses (opt-IN by configuring)        |
| Market-intelligence feed  | MARKET_FEED_URL, MARKET_FEED_TOKEN                    | advisor's market findings absent; own-data findings stand |
| LLM                       | AI_* (\$8)                                            | keyless stub answers                                      |
| PIM (product imagery)     | PIM_BASE_URL                                          | internal TMF667 store                                     |
| Web analytics (GA4)       | ANALYTICS_*                                           | internal insight only                                     |
| Social platform           | SOCIAL_API_URL, SOCIAL_ACCESS_TOKEN                   | no audience push / lead import                            |

Per-tenant variants ride suffixes: `_NOVA` , `_FJORD` , ... matching the placeholders in `tenants.yml` .



## 10 • Environment reference (dev clocks)

| VARIABLE                                             | MEANING                           | DEV / PROD<br>DEFAULT     |
|------------------------------------------------------|-----------------------------------|---------------------------|
| BSS_BILLING_DUNNING_TICK_MS / _GRACE_MS              | collections sweep /<br>grace      | 5s & 20s / 60s &<br>7d    |
| BSS_BILLING_DISTRIBUTION_TICK_MS /<br>_RETRY_SECONDS | delivery relay / backoff<br>base  | 3s & 4s / 30s &<br>60s    |
| BSS_SOM_COMMISSION_WINDOW_MS / _TICK_MS              | angrerett window /<br>harden tick | 15s & 5s / 14d & 1h       |
| BSS_SOM_TELESALES_EXPIRY_MS / _TICK_MS               | offer expiry / sweep              | 15s & 5s / 48h &<br>1h    |
| BSS_TENANTS_REFRESH_MS                               | live registry refresh             | 15s                       |
| BSS_GATEWAY_PARTNER_RATE_CAPACITY /<br>_WINDOW_MS    | per-partner rate limit            | 10 per 2s / 60 per<br>60s |

## 11 • Extension API reference (beyond the TMF standard surfaces)

| ENDPOINT                                                                   | WHO                                  | DOES                                     |
|----------------------------------------------------------------------------|--------------------------------------|------------------------------------------|
| GET /customerBill/{id}/document.pdf                                        | owner/staff                          | the bill as a PDF                        |
| POST /customerBill/{id}/resend                                             | owner/staff                          | email the invoice to the address on file |
| POST /customerBill/{id}/dispute · /installmentPlan                         | owner/agent                          | contest / split a bill                   |
| GET/PATCH/POST /billFormatProfile[/code]]                                  | read: billing · write: billing:admin | e-invoice profiles as rows               |
| GET /billDistribution · POST /billDistribution/{id}/retry                  | billing:admin                        | the delivery ledger                      |
| POST /bank/v1/remittance                                                   | the bank (X-Bank-Token)              | camt.054 / OCR / BAI2 in                 |
| POST /distribution/v1/response                                             | the partner (X-Distribution-Token)   | Peppol Invoice Response onto the ledger  |
| GET /remittance/unapplied                                                  | billing:admin                        | unapplied cash worklist                  |
| POST /individual/{id}/billingCycle · /billDelivery                         | self/staff                           | payday alignment · delivery preference   |
| /dealer/v1/* (agreement, kits, sell, commission, orders/{id}, telesales/*) | staff / dealer / partner POS         | the whole dealer & telesales channel     |
| POST /telesales/v1/confirm                                                 | the customer, signed in              | the written yes that births the order    |
| GET /advisor/v1/findings · POST /advisor/v1/adopt                          | catalog:write                        | product advisor · adopt-as-draft         |
| GET/POST /onboarding/v1/operator · PATCH .../{id}                          | host roles:admin                     | mint / live-mutate an operator           |
| GET /app/tenant-config.json                                                | public, by Host                      | the white-label manifest                 |

## 12 · The mock integrations (dev stand-ins)

| CONTAINER         | PORT | STANDS IN FOR                                                       | CHAOS / INSPECTION                                             |
|-------------------|------|---------------------------------------------------------------------|----------------------------------------------------------------|
| mock-esp          | 8121 | SendGrid-shaped email provider                                      | GET /mails , webhook receipts                                  |
| mock-social       | 8122 | Meta-shaped audiences/leads                                         | —                                                              |
| mock-distribution | 8124 | Peppol access point / print house (validates EHF/EDIFACT/Factur-X!) | GET /invoices , POST /outage , POST /invoices/{billNo}/respond |
| mock-dnc          | 8125 | reservation register                                                | numbers ending 9999 are reserved; POST /outage                 |
| mock-market       | 8126 | tariff-comparison feed                                              | GET /offers                                                    |
| mock-llm          | 8127 | OpenAI- and Anthropic-dialect LLM                                   | names the model in answers; GET /requests                      |

genalpha-bss · fifty-three suites, eleven CTKs at zero failures · the build story is *The Honest Machine* (docs/book) · the TMF vocabulary is docs/book/tmf-reference.md